Penumbra: day-ahead solar forecasts that know when they're unsure



Calibrated probabilistic forecasting for Belgium's 12.1 GW PV fleet, benchmarked against the grid operator — DataNova 2026 · Sustainability · DN-SU-01

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Live demo: https://s-kkipie.github.io/penumbra-solar-forecast/

Solar is now Belgium's largest day-ahead uncertainty

12.1 GW

installed PV, up from 9.1 GW

Elia monitored capacity, Apr 2024 → Oct 2026

335 MW

average hourly error of the TSO forecast

Elia D-1 11h, daylight hours, backtest period

1,539 MW

P99 shortfall to cover with reserve

hourly over-forecast, 99th percentile

- Day-ahead positions close at 12:00 on D-1: every MW of forecast error is bought back in balancing, mostly from gas plants.
- Point forecasts hide how sure they are. Reserves and trading need honest uncertainty — wide when skies are mixed, tight when clear.
- Who needs it: TSOs and DSOs (reserve sizing), balance-responsible parties and aggregators (bidding), and grids in emerging solar markets that lack a TSO-grade forecaster.

Three ideas, one daily forecast

1

Eight weather models, not one

ICON, GFS, ARPEGE/AROME, ECMWF IFS, HARMONIE (KNMI, DMI), GEM and ECMWF's AI model AIFS — combined per province and weighted by installed PV. Their disagreement is a feature.

2

Bands you can trust

LightGBM quantile regression (P10/P50/P90) + Mondrian conformal calibration per sun-elevation regime: the 80 % band covers ~80 % of hours at dawn, noon and dusk.

3

Proven in public

Hybrid mode also reads Elia's own 11:00 forecast (forecast combination). Every daily forecast is committed to git before market close and scored afterwards.

Everything runs on free public data and free infrastructure (GitHub Actions + GitHub Pages).

Real, public, verifiable data — 2.5 years, hourly

Source	What we use	Licence
Elia Open Data (ods032 / ods209 / ods210)	Measured PV output, installed capacity, 11 provinces, 15-min; Elia's D-1 11h forecast with P10/P90	Elia Open Data Licence
Open-Meteo Previous Runs API	8 NWP models archived at fixed 24 h / 48 h lead; GHI, direct, cloud, temperature at 11 provincial points	CC BY 4.0
Energy-Charts (Fraunhofer ISE) + Elia ods134	Day-ahead and imbalance prices — impact analysis only, never model input	CC BY 4.0 / Elia

- Quarter-hours → hours only when all four exist; numbers stored as text with decimal commas are parsed and validated.
- Radiation fields re-labelled to the interval they describe (verified: correlation peaks at zero lag).
- Installed capacity lagged 2 days; target = capacity factor, so 33 % fleet growth does not leak.

22,409

hourly targets, none imputed

0.88 % missing hours dropped

12,273

daylight hours used for scoring

sun elevation > −2°

Pipeline and the no-look-ahead rule

Ingest

Elia + 8 NWP models
11 provinces, hourly

Features

capacity-weighted irradiance, clear-sky index (pvlib), POA, model spread, spatial dispersion, trailing performance ratio

Model

LightGBM quantiles P10/P50/P90 (+ Elia 11h in Hybrid)

Calibrate

Mondrian conformal per sun-elevation regime, last 60 days

Deliver

GitHub Action 10:30 UTC → append-only log → dashboard

Honesty rules (unit-tested)

- Strict setting uses weather runs issued 48 h before each hour — all available before Elia's D-1 11:00 forecast.
- Rolling-origin backtest: refit every month, predict the next unseen month (19 months, 575 days).
- Every method scored on the identical set of daylight hours; significance by Diebold–Mariano + block bootstrap.

Ties the grid operator on accuracy, beats it on uncertainty

2.88 %

Hybrid nMAE (strict) — Elia 2.87 %

optimistic inputs: 2.74 % vs 2.87 %

29 %

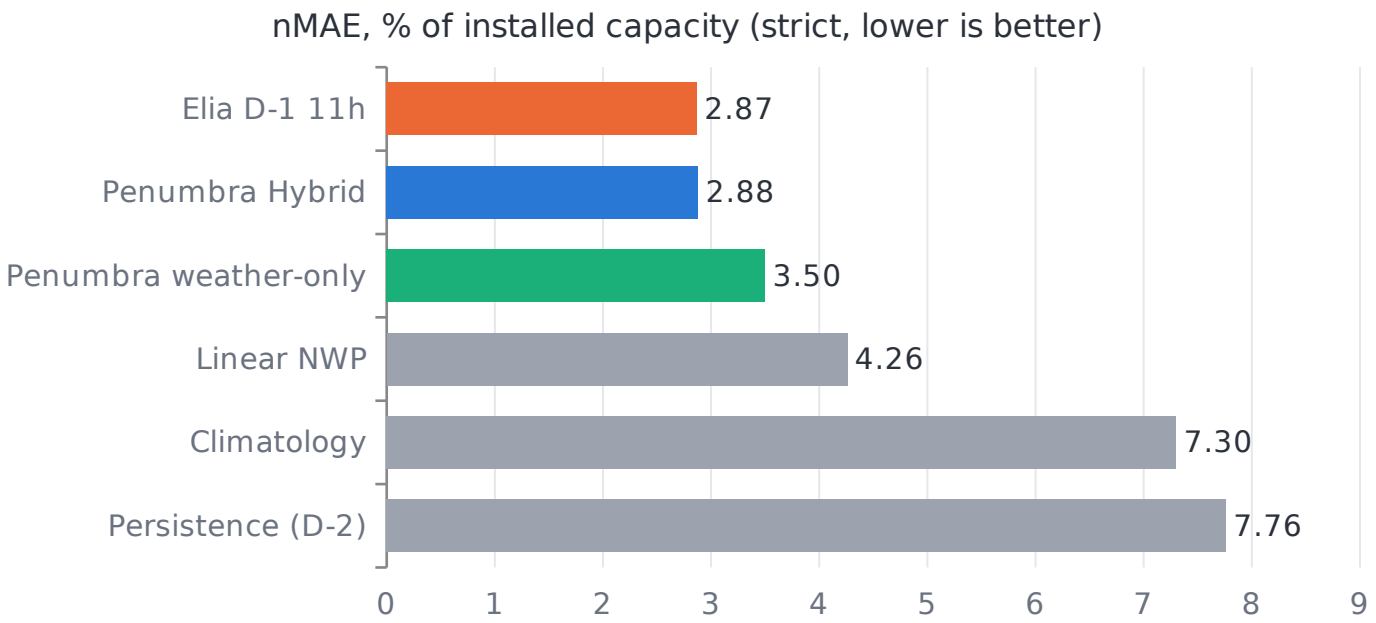
narrower 80 % band than Elia's P10–P90

coverage 81.2 % vs 80.0 %

10.4 %

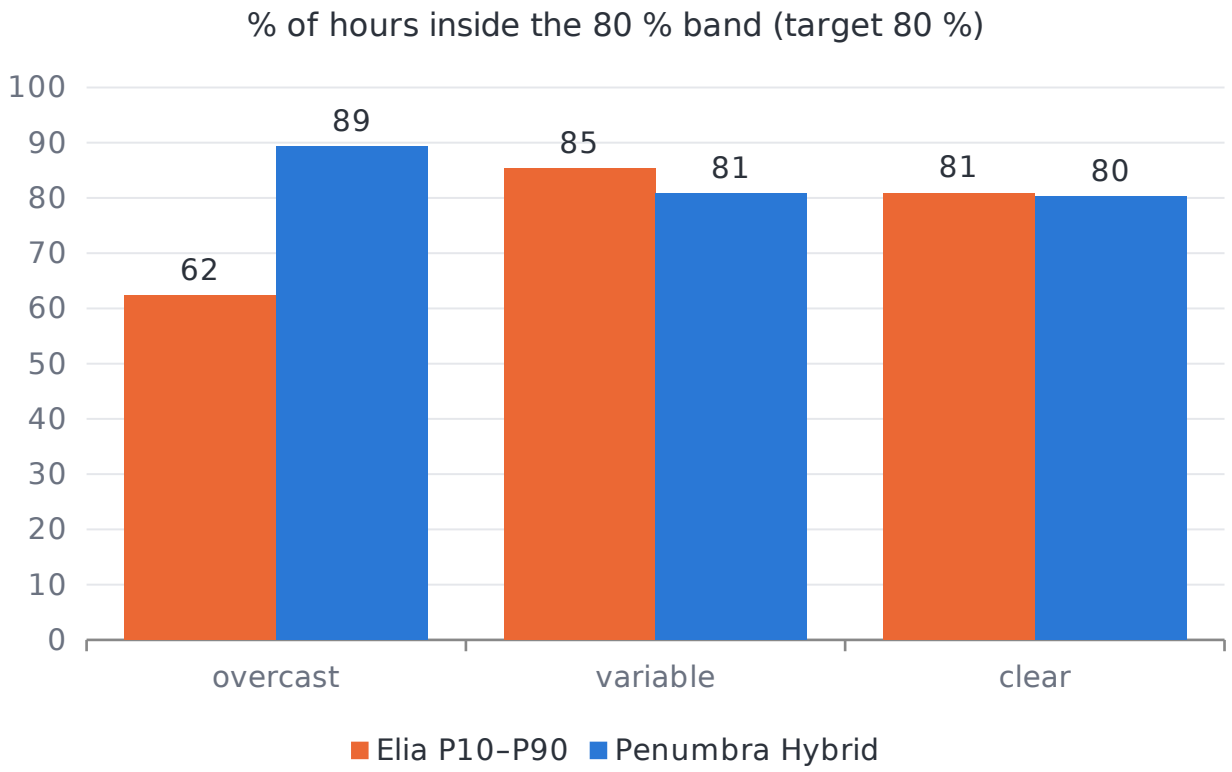
better Winkler interval score than Elia

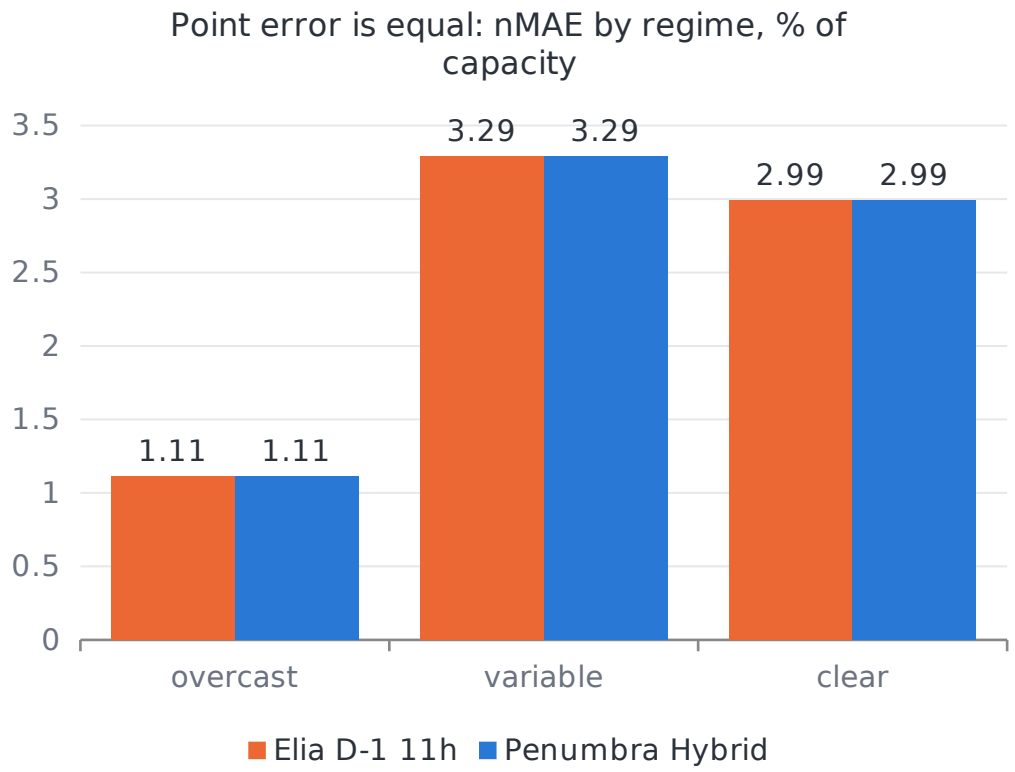
pinball 0.919 vs 0.969



- Weather-only Penumbra, from free data alone: 3.50 % (strict) / 3.08 % (optimistic) — 55 % better than persistence.
- Hybrid vs Elia on point error: −0.2 % (strict, DM p = 0.98) and +4.6 % (optimistic, p = 0.01).
- The gain is in uncertainty: same coverage, much tighter bands → less reserve held for the same risk.

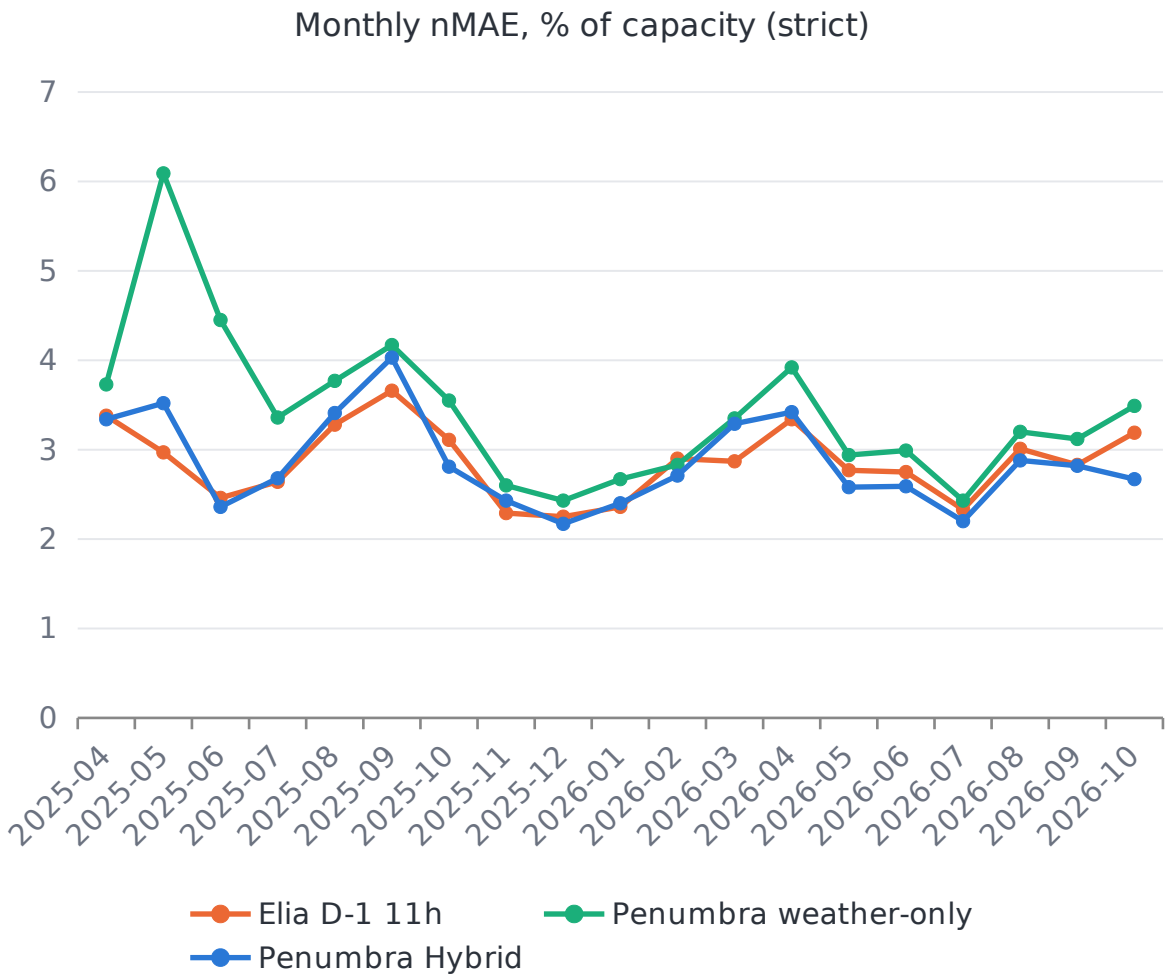
Calibration holds in every sky regime

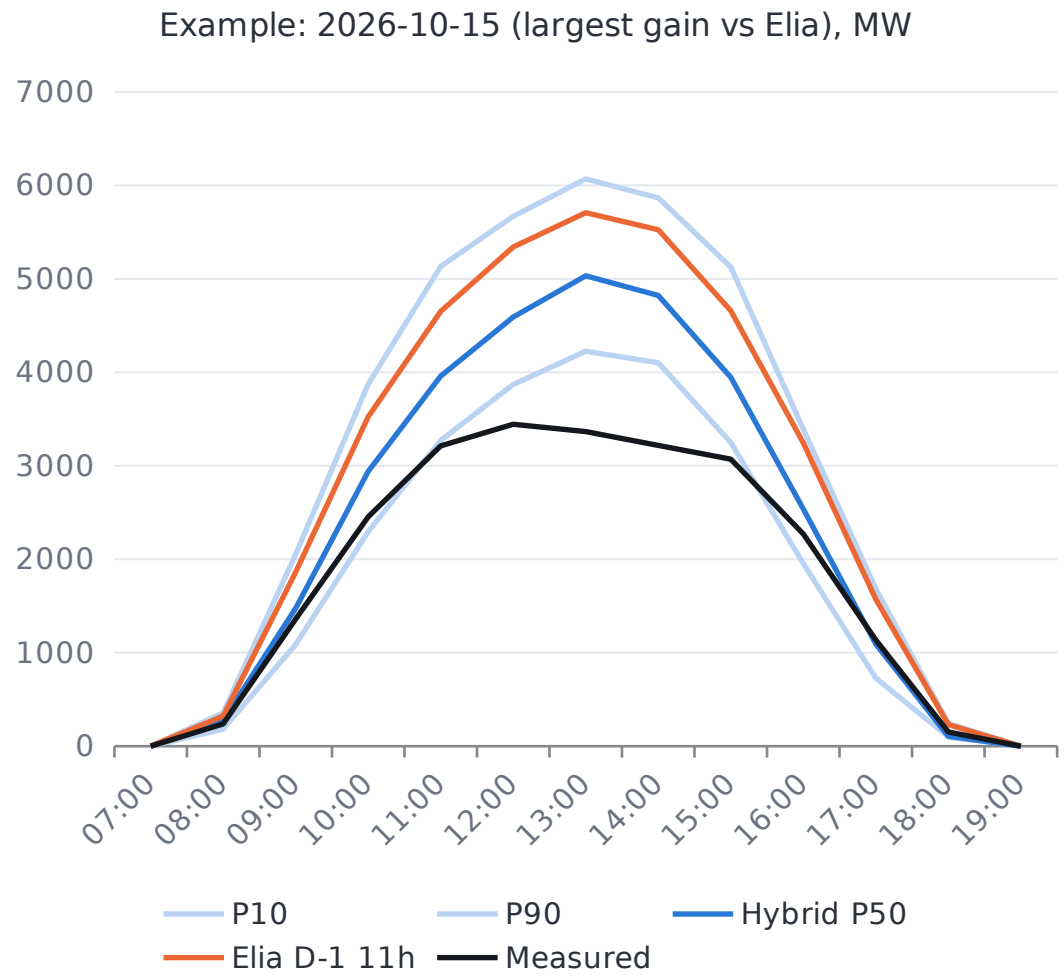




Regimes are defined from the forecast clear-sky index (known in advance). Elia's band is far too narrow when overcast and too wide when the sky is mixed; conformal calibration keeps Penumbra at or above 80 % in every regime while its average band is 8.9 % of capacity vs 12.5 %.

Stable across 19 out-of-sample months





What the improvement is worth

445 MW

narrower average 80 % band

at today's 12.1 GW, same coverage

7.45 €/MWh

imbalance cost of forecast error

Elia's forecast: 7.71 €/MWh

3.2 M€/yr

lower imbalance settlement

whole fleet, price-taker illustration

- Reserve sizing: a TSO dimensions upward/downward reserve from forecast uncertainty — a band that is narrower at equal coverage is directly less capacity held idle (mostly gas).
- Trading: a balance-responsible party can bid the quantile that minimises its expected imbalance cost instead of a single number.
- Honest caveat: the P99 hourly over-forecast is 79 MW worse than Elia's — concentrated in negative-price hours when PV is curtailed.
- Assumptions are explicit (price-taker, hourly settlement); figures are orders of magnitude, not a market simulation.

Feasible today, portable tomorrow

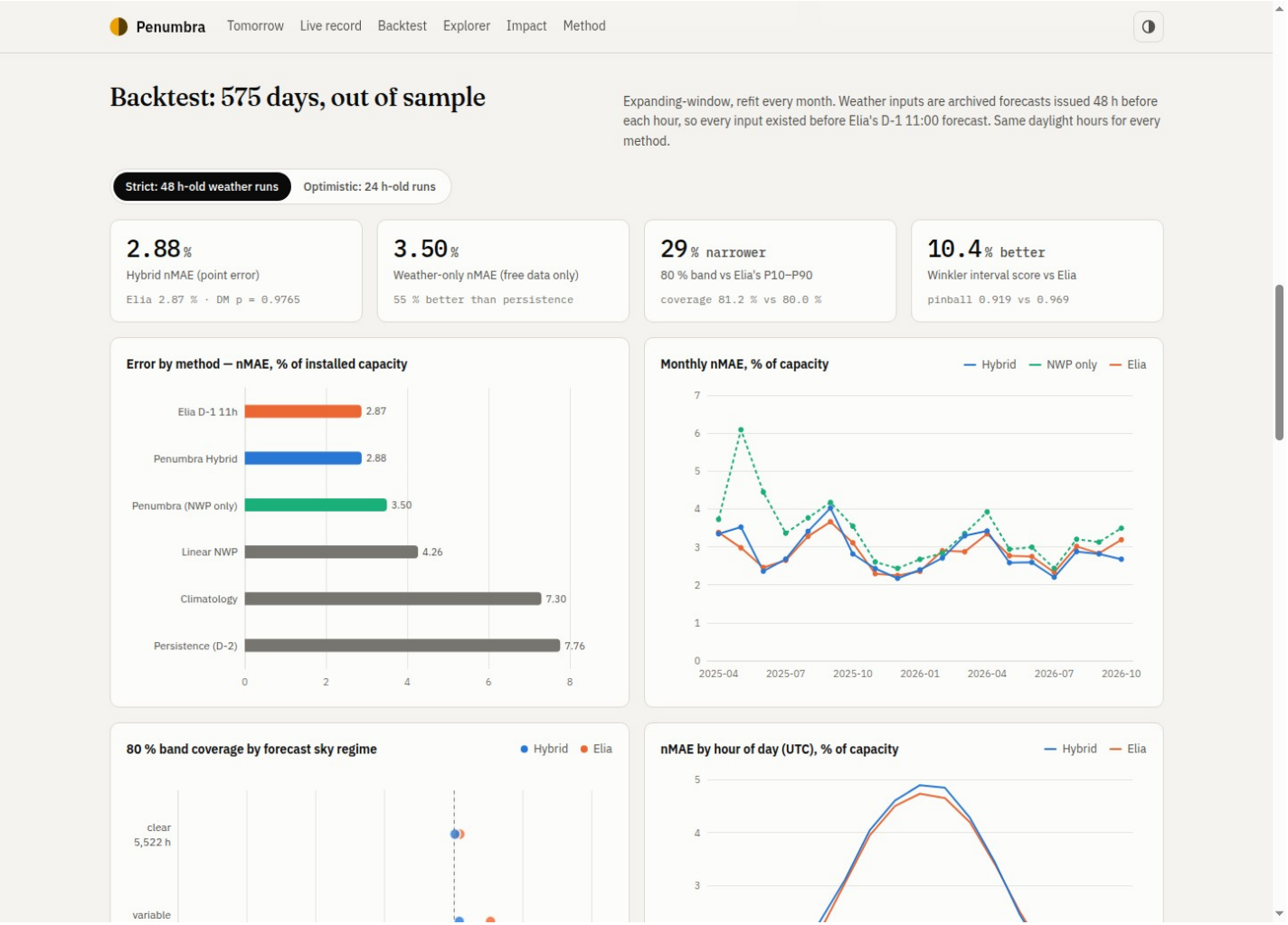
Runs in production now

- Daily GitHub Action issues tomorrow's forecast at 10:30 UTC, before the 12:00 CET gate.
- Append-only forecast log, scored automatically when Elia publishes actuals.
- Static dashboard on GitHub Pages: no server, no cost, no API keys.
- Full backtest reproduces in ~10 minutes on a laptop (download ~15 min).

Portable to data-scarce grids

- Weather-only mode needs just a production history and free global NWP — no TSO forecast.
- Peru's southern solar hub (Arequipa, Moquegua, Tacna) sits under the world's highest irradiance; COES dispatch data can replace Elia's.
- Capacity-factor target and per-region weighting carry over to any fleet size.
- Next: per-province forecasts for DSOs, intraday updates, curtailment-aware features from price forecasts.

Live dashboard and verifiable track record



- Tomorrow: fan chart with P10–P90 band next to Elia's forecast.
- Live record: daily error vs Elia, coverage, issue timestamps.
- Backtest: method comparison, monthly skill, calibration, significance tests.
- Explorer: any of the backtest days, with shortcuts to the hardest ones.

Demo

https://s-kkipie.github.io/penumbra-solar-forecast/

Code

https://github.com/S-kkipie/penumbra-solar-forecast

Limits, responsible use and team

Known limits

- Target is Elia's upscaled estimate of the fleet, not metered totals.
- Market-driven curtailment at negative prices is only proxied (weekend/holiday flag, trailing ratio).
- Accuracy depends on weather-run freshness: 48 h- vs 24 h-old runs cost 0.4 pp nMAE (weather-only).
- National aggregate only; per-province models are future work.

Responsible data use

Aggregate, non-personal public data under open licences; all sources credited; claims reproducible from the repo; no result is reported without its benchmark and confidence interval.

Team

Adrian Issac Mamani Quispe (Arequipa, Peru) — full-stack / AI engineer: problem framing, data pipeline, modelling, evaluation, live system, dashboard and documentation.